

What 2,148 government job postings actually say about pay

Five findings that survived the noise — pulled from a full NLP sector-classification pipeline, not just the headline numbers.

2,148

POSTINGS ANALYZED

8

SECTORS, ZERO-SHOT

72%

SENIOR VS ENTRY GAP

100%

PROCESSED OFFLINE

1**SECTOR PAY**

The smallest sector pays the most

Software & Technology holds only **50 of 2,148** postings (2.3%) — yet leads every sector on average pay (\$100,455). Scarcity, not headcount, is what NYC government pays a premium for in tech.

2**STRONGEST PREDICTOR**

Seniority beats sector, decisively

Senior roles average **\$139,145** against **\$80,814** at entry-level — a 72% gap. That single variable moves pay more than which of the 8 sectors a job sits in.

3**VOLUME ≠ VALUE**

The busiest sector ranks near the bottom on pay

Customer Support & Administration posts **855 jobs** — 42% of the entire dataset — but sits second-to-last in mean salary. Most applicants are competing in the highest-volume, lowest-leverage segment.

4**DATA CAREERS**

Data roles are a niche that already pays like a mid-tier sector

Just **38 postings** (1.88%) are data-related, but they average **\$93,642** — on par with Healthcare. The skill transfers into government roles that never carry a "Data" title.

5**SELF-AUDIT**

The \$274K "top job" is a ranking artifact, not a target

Titles like Professionalism Officer (\$274K) top the list only because they're single, senior, one-off postings. Sector averages — not individual title ceilings — are the honest benchmark, which is why this report leads with those instead.

The judgment calls behind the pipeline

Every stage was a decision, not a default — chosen for accuracy, for what the hardware could actually run, and for keeping the data on one machine.

CLASSIFICATION METHOD — THREE APPROACHES COMPARED BEFORE WRITING A LINE OF CODE

REJECTED

TF-IDF + KMeans
Fast and fully local, but keyword-only — it can't tell "engineer" in a software posting from "engineer" in a sanitation posting. Too brittle for messy civil-service titles.

REJECTED

Hosted LLM API
Best raw accuracy on paper — but every one of 2,148 postings would leave the machine and hit a third-party endpoint. Ruled out on data privacy alone, not cost.

CHOSEN

Local Sentence-Transformers (MiniLM)
Zero-shot semantic classification via weighted cosine similarity (title 0.7 / description 0.3) — near-LLM accuracy, runs offline on a laptop CPU, and no posting ever leaves local storage.



Zero data leakage, by design
Every stage — ingestion, cleaning, classification, feature engineering, analysis — ran against a local MySQL warehouse on the same machine. No posting, title, or salary figure was ever sent to an external API. The LLM route was evaluated and specifically rejected for this reason.

WHY EACH TOOL, AND NOT ANOTHER

spaCy (POS tagging)	Title cleaning — keeps NOUN/PROPN only, so it generalizes to titles never seen before instead of a fixed keyword list.
TF-IDF Vectorizer	Tested first — fastest and lightest on resources, but discarded for missing semantic/contextual meaning.
Local LLM (Ollama)	Tested for accuracy — too slow and resource-heavy for a full 2,148-row batch on this hardware.
SentenceTransformers	Final choice — the balance of accuracy, offline privacy, and speed the other two couldn't match together.

TECHNICAL WORK BEHIND THE NUMBERS

- › **4-stage SQL-backed ETL** — raw CSV → MySQL warehouse → NLP classification → feature engineering, every intermediate table persisted and auditable.
- › **5 engineered features** — Avg_Salary, Salary_Flexibility_%, Executive flag, Days_To_Apply, and a standardized 3-tier level scale from messy civil-service codes.
- › **Self-audited the results** — flagged mean-vs-median skew and recommended ANOVA/Kruskal-Wallis and correlation testing before treating any pattern as confirmed.
- › **Withheld predictive modeling on purpose** — strata as small as n=1 (Senior data roles) can't support a trained model without curve-fitting to noise.